

Successive Differentiation

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Defⁿ:

The function f' defined by the formula

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

is called the derivative of f with respect to x .

The process of finding a derivative is called the differentiation. The differentiation operation is often denoted by the symbol $\frac{d}{dx} [f(x)]$. Thus $\frac{d}{dx} [f(x)] = f'(x)$.

The derivative $f'(x)$ gives the slope of the graph at x and also the instantaneous rate of change of f at x .

Differentials:

Let $\Delta x = dx$ be an increment gives to x , then

$\Delta y = f(x + \Delta x) - f(x)$ is called the increment in

$y = f(x)$. If $f(x)$ is continuous and a continuous first derivative in an interval, then

$$\Delta y = f'(x) \Delta x + \epsilon \cdot \Delta x$$

$$= f'(x) \Delta x + \epsilon dx \quad \text{--- (2)}$$

when $\epsilon \rightarrow 0$ as $\Delta x \rightarrow 0$

The expression $dy = f'(x) dx$ ——— ③ is called the differential of $y = f(x)$.

The quantity dx , called the differential of x and dy need not be small.

Because of definitions ① and ③, we often write

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

It is emphasized that dx and dy are not the limits of Δx and Δy as $\Delta x \rightarrow 0$. Instead, dy is a dependent variable determined from the independent variable dx for a given x .

Successive Differentiation

Some basic theorems:

Th^m:

If $y = x^n$, $n \in \mathbb{N}$, then show that $y_n = n!$

Proof:

Given, $y = x^n$

Differentiating wrt x we get

$$y_1 = nx^{n-1}$$

Again differentiating wrt x we obtain

$$y_2 = n(n-1)x^{n-2}$$

Similarly,

$$y_3 = n(n-1)(n-2)x^{n-3}$$

$$y_n = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 x^{n-n}$$

$$= n(n-1)(n-2) \dots 2 \cdot 1$$

$$= n!$$

(Proved)

Th^m:

If $y = x^m$ and $m \in \mathbb{N}$, then show that

(a) $y_n = 0$, when $n > m$

(b) $y_n = \frac{m!}{(m-n)!} x^{m-n}$, when $n \leq m$.

Proof: (a)

Given, $y = x^m$

Therefore, using the previous th^m we obtain

$$y_m = m!$$

Thus, $y_n = 0$ if $n > m$.

Proof: (b)

Given, $y = x^m$

Diff w.r.t x we get

$$y_1 = mx^{m-1}$$

Diff. again w.r.t x we obtain

$$y_2 = m(m-1)x^{m-2}$$

Similar

Similarly,

$$y_3 = m(m-1)(m-2)x^{m-3}$$

$$y_n = m(m-1)(m-2)\dots\{m-(n-1)\}x^{m-n} \text{ when } m \geq n$$

$$\Rightarrow y_n = m(m-1)(m-2)\dots(m-n+1)x^{m-n}$$

$$= \frac{m(m-1)(m-2)\dots(m-n+1)(m-n)!}{(m-n)!} x^{m-n}$$

$$= \frac{m(m-1)(m-2)\dots(m-n+1)(m-n)(m-n-1)\dots 3 \cdot 2 \cdot 1}{(m-n)!} x^{m-n}$$

$$= \frac{m!}{(m-n)!} x^{m-n}, \text{ when } m \geq n$$

(Proved)

Th^m:

If $y = (ax+b)^n$ and $n \in \mathbb{N}$, then show that $y_n = n! a^n$

Proof:

Given, $y = (ax+b)^n$

Diff. wrt x we get.

$$y_1 = n(ax+b)^{n-1} \cdot a$$

Again diff wrt x we have

$$y_2 = n(n-1)(ax+b)^{n-2} \cdot a^2$$

Similarly,

$$y_3 = n(n-1)(n-2)(ax+b)^{n-3} a^3$$

$$\vdots$$

$$y_n = n(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1 \cdot a^n$$

$$\therefore y_n = n! a^n$$

(Proved)

Th^m:

If $y = a^{bx+c}$, then show that $y_n = a^{bx+c} (\ln a)^n$

Proof:

$$\text{Given, } y = a^{bx+c} \quad \text{--- ①}$$

Diff ① wrt x we get

$$y_1 = a^{bx+c} \cdot \ln a \cdot b$$

Again diff wrt x we have

$$y_2 = a^{bx+c} (\ln a)^2 \cdot b$$

Simil

Similarly,

$$y_3 = e^{bx+c} (\ln a)^3 b^3$$

$$y_n = e^{bx+c} (\ln a)^n b^n$$

(Proved) $\rightarrow$

Th^m:

If $y = \ln(ax+b)$, then show that

$$y_n = (-1)^{n-1} (n-1)! (ax+b)^{-n} a^n$$

Proof:

Given,

$$y = \ln(ax+b) \quad \text{--- (1)}$$

Diff. (1) wrt x we get.

$$y_1 = \frac{1}{ax+b} \cdot a = (ax+b)^{-1} \cdot a$$

Again diff wrt x we have

$$y_2 = (-1) (ax+b)^{-2} \cdot a^2$$

Similarly,

$$y_3 = (-1)(-2) (ax+b)^{-3} \cdot a^3$$

$$y_n = (-1)(-2) \dots (-n+1)(ax+b)^{-n} \cdot a^n$$

$$= (-1)^{n-1} [1 \cdot 2 \cdot 3 \dots (n-1)] (ax+b)^{-n} a^n$$

$$= (-1)^{n-1} (n-1)! (ax+b)^{-n} a^n$$

(Proved)

Th^m:

If $y = \sin(ax+b)$, then show that

$$y_n = a^n \sin\left(n \cdot \frac{\pi}{2} + ax+b\right)$$

Proof:

Given, $y = \sin(ax+b)$ ————— ①

Diff ① w.r.t x , we have

$$y_1 = a \cos(ax+b) = a \sin\left(\frac{\pi}{2} + ax+b\right)$$

Diff. again w.r.t x we get.

$$\begin{aligned} y_2 &= a^2 \cos\left(\frac{\pi}{2} + ax+b\right) = a^2 \sin\left(\frac{\pi}{2} + \frac{\pi}{2} + ax+b\right) \\ &= a^2 \sin\left(2 \cdot \frac{\pi}{2} + ax+b\right) \end{aligned}$$

Similarly,

$$y_3 = a^3 \cos\left(2 \cdot \frac{\pi}{2} + ax+b\right) = a^3 \sin\left(\frac{\pi}{2} + 2 \cdot \frac{\pi}{2} + ax+b\right)$$

$$\Rightarrow y_3 = a^3 \sin\left(3 \cdot \frac{\pi}{2} + ax + b\right)$$

$$y_n = a^n \sin\left(n \cdot \frac{\pi}{2} + ax + b\right)$$

(Proved)

Ex^m:

If $y = \cos(ax + b)$, then show that

$$y_n = a^n \cos\left(n \cdot \frac{\pi}{2} + ax + b\right)$$

Proof:

Given, $y = \cos(ax + b)$ ——— ①

Diff ① w.r.t x we get.

$$y_1 = -a \sin(ax + b) = a \cos\left(\frac{\pi}{2} + ax + b\right)$$

Again diff w.r.t x we have

$$y_2 = -a^2 \sin\left(\frac{\pi}{2} + ax + b\right) = a^2 \cos\left(\frac{\pi}{2} + \frac{\pi}{2} + ax + b\right)$$

$$\Rightarrow y_2 = a^2 \cos\left(2 \cdot \frac{\pi}{2} + ax + b\right)$$

Similarly, $y_3 = -a^3 \sin\left(2 \cdot \frac{\pi}{2} + ax + b\right)$

$$\Rightarrow y_3 = a^3 \cos\left(\frac{\pi}{2} + 2 \cdot \frac{\pi}{2} + ax + b\right)$$

$$\Rightarrow y_3 = a^3 \cos\left(3 \cdot \frac{\pi}{2} + ax + b\right)$$

$$y_n = a^n \cos\left(n \cdot \frac{\pi}{2} + ax + b\right)$$

(Proved)

Th^m :

If $y = e^{ax} \sin(bx + c)$, then show that

$$y_n = (a^2 + b^2)^{n/2} e^{ax} \sin(bx + c + n \tan^{-1} \frac{b}{a})$$

Proof:

Given, $y = e^{ax} \sin(bx + c)$ ————— ①

Differentiating ① wrt x we get

$$y_1 = a e^{ax} \sin(bx + c) + b e^{ax} \cos(bx + c)$$

$$\Rightarrow y_1 = e^{ax} [a \sin(bx + c) + b \cos(bx + c)] \text{ ————— ②}$$

Let, $a = r \cos \theta$ & $b = r \sin \theta$

Hence, $a^2 + b^2 = r^2 (\cos^2 \theta + \sin^2 \theta) = r^2$

$$\Rightarrow r = (a^2 + b^2)^{1/2}$$

$$\& \frac{b}{a} = \tan \theta$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

Now, from eqnⁿ (2) we get

$$y_1 = e^{ax} \left[r \cos \theta \sin(bx+c) + r \sin \theta \cos(bx+c) \right]$$

$$= r e^{ax} \left[\cos \theta \sin(bx+c) + \sin \theta \cos(bx+c) \right]$$

$$\therefore y_1 = r e^{ax} \sin(bx+c+\theta)$$

Similarly, differentiating y_1 w.r.t x we obtain

$$y_2 = r^2 e^{ax} \sin(bx+c+2\theta)$$

Proceeding in the similar manner, we have

$$y_n = r^n e^{ax} \sin(bx+c+n\theta)$$

$$\therefore y_n = (a^2 + b^2)^{n/2} e^{ax} \sin(bx+c+n \tan^{-1}\left(\frac{b}{a}\right))$$

(Proved)

Th^m:

If $y = e^{ax} \cos(bx+c)$, then show that

$$y_n = (a^2 + b^2)^{n/2} e^{ax} \cos(bx+c+n \tan^{-1} \frac{b}{a}).$$

Proof:

Given, $y = e^{ax} \cos(bx+c)$ _____ ①

Differentiating ① wrt x we have

$$y_1 = ae^{ax} \cos(bx+c) - be^{ax} \sin(bx+c)$$

$$\Rightarrow y_1 = e^{ax} [a \cos(bx+c) - b \sin(bx+c)] \text{ --- ②}$$

Let, $a = r \cos \theta$ & $b = r \sin \theta$

so that $a^2 + b^2 = r^2 \Rightarrow r = (a^2 + b^2)^{1/2}$

$$\tan \theta = \frac{a}{b} \Rightarrow \theta = \tan^{-1} \left(\frac{a}{b} \right)$$

Hence, from ② we have

$$y_1 = e^{ax} [r \cos \theta \cos(bx+c) - r \sin \theta \sin(bx+c)]$$

$$\Rightarrow y_1 = re^{ax} [\cos \theta \cos(bx+c) - \sin \theta \sin(bx+c)]$$

$$\Rightarrow y_1 = re^{ax} \cos(bx+c+\theta)$$

Similarly

Similarly, differentiating y_1 w.r.t x we have

$$y_2 = r^n e^{ax} \cos(bx + c + 2\theta)$$

Proceeding in a similar manner, we obtain

$$y_n = r^n e^{ax} \cos(bx + c + n\theta)$$

$$\therefore y_n = (a^2 + b^2)^{n/2} e^{ax} \cos(bx + c + n \tan^{-1}(b/a))$$

(Proved)

Example:

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If $y = x^{2n}$ then show that $y_n = 2^n \{1 \cdot 3 \cdot 5 \cdots (2n-1)\} x^n$ where $n \in \mathbb{N}$.

Solⁿ:

Given, $y = x^{2n}$ ———— (1)

Diff (1) w.r.t x we have

$$y_1 = 2nx^{2n-1}$$

Again diff w.r.t x we have

$$y_2 = 2n(2n-1)x^{2n-2}$$

$$y_3 = 2n(2n-1)(2n-2)x^{2n-3}$$

$$y_4 = 2n(2n-1)(2n-2)(2n-3)x^{2n-4}$$

$$\begin{aligned}
 Y_n &= 2n(2n-1)(2n-2)(2n-3) \dots \{2n-(n-1)\} x^{2n} \\
 &= 2n(2n-1)(2n-2)(2n-3) \dots (n+1) x^n \\
 &= \frac{2n(2n-1)(2n-2) \dots (n+1) \cdot n!}{n!} x^n
 \end{aligned}$$

$$\therefore Y_n = \frac{(2n)!}{n!} x^n \quad \text{--- (2)}$$

Now,

$$\begin{aligned}
 (2n)! &= 2n(2n-1)(2n-2)(2n-3)(2n-4) \dots 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\
 &= \{2n(2n-2)(2n-4) \dots 6 \cdot 4 \cdot 2\} \{(2n-1)(2n-3) \dots 5 \cdot 3\} \\
 &= 2^n \{n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1\} \{(2n-1)(2n-3) \dots 5 \cdot 3 \cdot 1\} \\
 &= (2^n \cdot n!) \cdot \{(2n-1)(2n-3) \dots 5 \cdot 3 \cdot 1\}
 \end{aligned}$$

$$\Rightarrow \frac{(2n)!}{n!} = 2^n \{(2n-1)(2n-3) \dots 5 \cdot 3 \cdot 1\} \quad \text{--- (3)}$$

Now, using (3) eqn (2) can be rewritten as

$$Y_n = 2^n \{1 \cdot 3 \cdot 5 \dots (2n-3)(2n-1)\} x^n ; n \in \mathbb{N}.$$

(Shown)

Problem:

If $y = \sin(\sin x)$, then show that

$$\frac{d^2 y}{dx^2} + \tan x \frac{dy}{dx} + y \cos^3 x = 0$$

Solⁿ:

Given, $y = \sin(\sin x)$ ——— ①

Diff. ① w.r.t x we have

$$\frac{dy}{dx} = \cos(\sin x) \cos x \text{ ——— ②}$$

Again diff ② w.r.t x we have

$$\frac{d^2 y}{dx^2} = -\sin(\sin x) \cos^2 x - \cos(\sin x) \sin x$$

$$= -y \cos^2 x - \frac{1}{\cos x} \frac{dy}{dx} \cdot \sin x \left[\begin{array}{l} \text{using ①} \\ \& \text{②} \end{array} \right]$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -y \cos^2 x - \tan x \frac{dy}{dx}$$

$$\Rightarrow \frac{d^2 y}{dx^2} + \tan x \frac{dy}{dx} + y \cos^2 x = 0$$

(Shown)

Problem:

If $y = \ln(x + \sqrt{a^2 + x^2})$ then show that
 $(a^2 + x^2)y_2 + xy_1 = 0$

Solⁿ:

Given, $y = \ln(x + \sqrt{a^2 + x^2})$ ——— ①

Diff ① wrt x we have

$$y_1 = \frac{1}{x + \sqrt{a^2 + x^2}} \left(1 + \frac{1}{2\sqrt{a^2 + x^2}} \cdot 2x \right)$$

$$= \frac{1}{x + \sqrt{a^2 + x^2}} \left(1 + \frac{x}{\sqrt{a^2 + x^2}} \right)$$

$$= \frac{1}{(x + \sqrt{a^2 + x^2})} \left[\frac{(x + \sqrt{a^2 + x^2})}{\sqrt{a^2 + x^2}} \right]$$

$$\Rightarrow y_1 = \frac{1}{\sqrt{a^2 + x^2}} = (a^2 + x^2)^{-1/2} \text{ ——— ②}$$

Again diff ② wrt x we obtain

$$y_2 = -\frac{1}{2} (a^2 + x^2)^{-3/2} \cdot 2x$$

$$\Rightarrow y_2 = -x (a^2 + x^2)^{-1} \cdot (a^2 + x^2)^{-1/2}$$

$$\Rightarrow (a^2 + x^2)y_2 = -x (a^2 + x^2)^{-1/2} = -xy_1 \text{ [By ②]}$$

$$\Rightarrow (a^2 + x^2) y_2 + x y_1 = 0$$

(Showed)

Problem :

If $y = a \cos(\ln x) + b \sin(\ln x)$ then show that

$$x^2 y_2 + x y_1 + y = 0$$

Solⁿ :

Given, $y = a \cos(\ln x) + b \sin(\ln x)$ ———— ①

Diff ① w.r.t x we have

$$y_1 = -a \sin(\ln x) \cdot \frac{1}{x} + b \cos(\ln x) \cdot \frac{1}{x}$$

$$\Rightarrow x y_1 = -a \sin(\ln x) + b \cos(\ln x)$$
 ———— ②

Again diff ② w.r.t x we have

$$y_1 + x y_2 = -a \cos(\ln x) \cdot \frac{1}{x} - b \sin(\ln x) \cdot \frac{1}{x}$$

$$\Rightarrow x y_1 + x^2 y_2 = -[a \cos(\ln x) + b \sin(\ln x)]$$

$$\Rightarrow x^2 y_2 + x y_1 = -y \quad [\text{using ①}]$$

$$\Rightarrow x^2 y_2 + x y_1 + y = 0$$

(Showed)

Problem:

If $y = \sin(m \sin^{-1} x)$ then show that

$$(1-x^2)y_2 - xy_1 + m^2 y = 0.$$

Soln:

Given, $y = \sin(m \sin^{-1} x)$ ————— (1)

Diff (1) w.r.t x we have

$$y_1 = \cos(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 \sqrt{1-x^2} = m \cos(m \sin^{-1} x) \text{ ————— (2)}$$

Again diff (2) w.r.t x we obtain

$$y_2 \sqrt{1-x^2} + y_1 \cdot \frac{1}{\sqrt{1-x^2}} (-x) = -m \sin(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow y_2 (1-x^2) - xy_1 = -m^2 \sin(m \sin^{-1} x)$$

$$\Rightarrow (1-x^2)y_2 - xy_1 + m^2 y = 0 \quad [\text{using (1)}]$$

(Showed)

Problem:

If $\ln y = \tan^{-1} x$ then show that

$$(1+x^2)y_2 + (2x-1)y_1 = 0$$

Solⁿ:

Given, $\ln y = \tan^{-1} x$ ——— ①

Diff ① wrt x we have

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = y \text{ ——— ②}$$

Again diff ② wrt x we obtain

$$(1+x^2) y_2 + 2xy_1 = y_1$$

$$\Rightarrow (1+x^2) y_2 + (2x-1) y_1 = 0$$

(Showed)

Problem:

If $y = e^{\cot^{-1} x}$ then show that $(1+x^2) y_2 + (2x+1) y_1 = 0$

Solⁿ:

Given, $y = e^{\cot^{-1} x}$ ——— ①

Diff ① wrt x we obtain

$$y_1 = e^{\cot^{-1} x} \left(-\frac{1}{1+x^2} \right)$$

$$\Rightarrow (1+x^2) y_1 = -e^{\cot^{-1} x} = -y \text{ [using ①]}$$

$$\Rightarrow (1+x^2) y_1 + y = 0 \text{ ——— ②}$$

Again diff ② wrt x we obtain

$$(1+x^2) y_2 + 2xy_1 + y_1 = 0$$

$$\Rightarrow (1+x^2) y_2 + (2x+1) y_1 = 0$$

(shown)

Problem:

If $y = (\sin^{-1} x)^2$ then show that $(1-x^2) y_2 - xy_1 - 2 = 0$

Sol:

Given, $y = (\sin^{-1} x)^2$ ———— (1)

Diff (1) w.r.t x we obtain

$$y_1 = 2 \sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 \sqrt{1-x^2} = 2 \sin^{-1} x \text{ ———— (2)}$$

Again diff (2) w.r.t x we have

$$y_2 \sqrt{1-x^2} + y_1 \frac{1}{\cancel{\sqrt{1-x^2}}} (-\cancel{\sqrt{1-x^2}}) = 2 \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_2 (1-x^2) - xy_1 = 2$$

$$\Rightarrow (1-x^2) y_2 - xy_1 - 2 = 0$$

(shown)

Problem:

If $y = (x^2 - 1)^n$ then show that $(x^2 - 1)y_2 + 2(1 - n)xy_1 - 2ny = 0$

Solⁿ:

Given, $y = (x^2 - 1)^n$ ——— ①

Diff ① wrt x we obtain

$$y_1 = n(x^2 - 1)^{n-1} \cdot 2x$$

$$\Rightarrow y_1 = \frac{2nx(x^2 - 1)^n}{(x^2 - 1)}$$

$$\Rightarrow (x^2 - 1)y_1 = 2nx(x^2 - 1)^n = 2nxy \quad [\text{using ①}]$$

————— ②

Again diff ② wrt x we have

$$(x^2 - 1)y_2 + 2xy_1 = 2ny + 2nxy_1$$

$$\Rightarrow (x^2 - 1)y_2 - 2nxy_1 + 2xy_1 - 2ny = 0$$

$$\Rightarrow (x^2 - 1)y_2 - 2(n - 1)xy_1 - 2ny = 0$$

$$\Rightarrow (x^2 - 1)y_2 + 2(1 - n)xy_1 - 2ny = 0$$

(Showed)

Problem:

If $y = e^{a \sin^{-1} x}$ then show that $(1-x^2)y_2 - xy_1 - a^2 y = 0$

Sol:

Given, $y = e^{a \sin^{-1} x}$ ——— ①

Diff ① w.r.t x we get .

$$y_1 = e^{a \sin^{-1} x} \cdot \frac{a}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} \cdot y_1 = a e^{a \sin^{-1} x} = ay \text{ ——— ② [using ①]}$$

Again diff ② w.r.t x we obtain

$$(\sqrt{1-x^2}) y_2 + y_1 \cdot \frac{1}{\sqrt{1-x^2}} \cdot (-x) = ay_1$$

$$\Rightarrow (1-x^2) y_2 - xy_1 = a y_1 \sqrt{1-x^2}$$

$$\Rightarrow (1-x^2) y_2 - xy_1 = a \cdot ay \text{ [using ②]}$$

$$\Rightarrow (1-x^2) y_2 - xy_1 = a^2 y$$

$$\Rightarrow (1-x^2) y_2 - xy_1 - a^2 y = 0$$

(Showered)

Problem:

If $y = \sin(m \sin^{-1} x)$ then show that

$$(1-x^2) y_3 - 3xy_2 + (m^2-1) y_1 = 0$$

Soln:

Given, $y = \sin(m \sin^{-1} x)$ ——— ①

Diff ① wrt x we have

$$y_1 = \cos(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 \sqrt{1-x^2} = m \cos(m \sin^{-1} x) \text{ ——— ②}$$

Again diff ② wrt x we obtain

$$y_2 \sqrt{1-x^2} + y_1 \frac{1}{\cancel{2}\sqrt{1-x^2}} (-2x) = -m \sin(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow (\sqrt{1-x^2}) y_2 - xy_1 \cdot \frac{1}{\sqrt{1-x^2}} = -\frac{m^2 y}{\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2) y_2 - xy_1 = -m^2 y \text{ ——— ③}$$

Diff ③ wrt x gives

$$(1-x^2) y_3 - 2xy_2 - y_1 - xy_2 = -m^2 y_1$$

$$\Rightarrow (1-x^2) y_3 - 3xy_2 + (m^2-1) y_1 = 0, \text{ showed.}$$

Problem:

If $y = x^2 \ln x$ then show that $xy_3 = 2$.

Solⁿ:

Given, $y = x^2 \ln x$ ——— ①

Diff ① w.r.t x we obtain

$$y_1 = 2x \ln x + x^2 \cdot \frac{1}{x}$$

$$\Rightarrow y_1 = 2x \ln x + x \text{ ——— ②}$$

Again diff ② w.r.t x we have

$$y_2 = 2x \cdot \frac{1}{x} + 2 \ln x + 1$$

$$\Rightarrow y_2 = 2 + 2 \ln x + 1 = 2 \ln x + 3 \text{ ——— ③}$$

Again diff ③ w.r.t x we get

$$y_3 = 2 \cdot \frac{1}{x}$$

$$\Rightarrow xy_3 = 2$$

(shown)

Problem:

If $\ln y = \tan^{-1} x$ then show that

$$(1+x^2) y_3 + (4x-1) y_2 + 2y_1 = 0$$

Soln:

$$\text{Given, } \ln y = \tan^{-1} x \quad \text{--- (1)}$$

Diff (1) wrt x we get

$$\frac{1}{y} y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = y \quad \text{--- (2)}$$

Diff (2) wrt x we obtain

$$(1+x^2) y_2 + 2xy_1 = y_1$$

$$\Rightarrow (1+x^2) y_2 + (2x-1) y_1 = 0 \quad \text{--- (3)}$$

Diff (3) wrt x we have

$$(1+x^2) y_3 + 2xy_2 + 2y_1 + (2x-1) y_2 = 0$$

$$\Rightarrow (1+x^2) y_3 + (4x-1) y_2 + 2y_1 = 0$$

(Showered)